Team Name : Astro4

Team Leader Name : Kartik Sisodia

Problem Statement 3: Development of Surface AQI & Identification of HCHO Hotspots over India using Satellite Data

Team Members

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BRIEF ABOUT THE IDEA

HOW OUR SOLUTION SOLVES THE PROBLEM

- Combines satellite data, weather data and ground measurements to understand the relationship between atmospheric conditions and surface level pollution.
- Uses an optimized XGBoost model to predict 6 major pollutants ($PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3) simultaneously and also supports HCHO analysis.
- Creates pollution maps and identifies hotspots so that areas with less monitoring stations can also be tracked continuously.
- Helps in HCHO hotspot detection, spatio-temporal trend analysis and atmospheric transport assessment for better pollution characterization.

HOW IS IT DIFFERENT FROM EXISTING IDEAS?

- Not limited to only ground stations or single satellite products. We fuse INSAT-3DR, ERA5 and CPCB data for better coverage.
- Instead of using a fixed area, we test multiple AOD neighbourhoods (1x1, 3x3, 5x5) and use the one that gives the best results.
- Dual objective approach: along with predicting 6 pollutants, we also do HCHO hotspot detection, time trend analysis and transport study in one pipeline.
- One unified AI framework for multi-pollutant prediction, hotspot detection and transport analysis.

OUR APPROACH PIPELINE

1. **INSAT-3DR Satellite Data** (AOD, HCHO, Trace Gases) →
2. **ERA5 Meteorological Data** (Temperature, RH, Wind, Rainfall) →
3. **CPCB Ground Observations** ($PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3) →
4. **Data Preprocessing & Feature Engineering** →
5. **XGBoost Multi-Output Model** (Predicts 6 Pollutants) →

Objective 1
Multi-Pollutant Prediction
($PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3)

Objective 2
HCHO Hotspot Analysis & Transport Study

KEY INNOVATIONS

- **Evidence-Based Model Selection**: Evaluated CNN-LSTM, Random Forest and XGBoost. Selected XGBoost for best accuracy and efficiency.
- **Multi-Scale AOD Optimization**: Tested 1x1, 3x3 and 5x5 INSAT-3DR AOD windows and selected the best representation.
- **Lightweight & Scalable**: High accuracy ($R^2 \approx 0.84$) with a compact model that can be deployed quickly.
- **Unified Air Quality Intelligence**: One AI model for prediction + HCHO hotspot + transport analysis.

LIST OF FEATURES

1. Multi-Pollutant Prediction

- Simultaneous estimation of six major pollutants: $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, and O_3.
- Utilizes INSAT-3DR satellite products (AOD, HCHO, Gases), ERA5 meteorological data (Temperature, RH, Wind, PBL, Rainfall), and CPCB ground observations.
- XGBoost multi-output regression model captures complex, non-linear relationships between atmospheric variables and ground-level pollutant concentrations.
- Delivers accurate hourly concentration forecasts across regions and time.

2. HCHO Hotspot Intelligence

- Performs high-resolution HCHO concentration mapping using INSAT-3DR HCHO data.
- Identifies hotspots and high-intensity regions through spatial clustering (e.g., DBSCAN) and intensity thresholding.
- Analyzes temporal trends to monitor hotspot evolution and detect persistent sources.
- Supports emission-prone region identification and aids in targeted pollution control strategies.

3. Multi-Scale AOD Optimization

- Systematically evaluates 1×1 (point), 3×3, and 5×5 INSAT-3DR AOD neighbourhoods.
- Compares spatial representations to determine the optimal scale for pollution estimation.
- Experimental evaluation identified the 5×5 neighbourhood as the optimal representation, achieving the best predictive performance.
- Enhances model robustness by capturing surrounding spatial context.

4. Multi-Source Atmospheric Data Fusion

- Integrates heterogeneous data sources: INSAT-3DR satellite products, ERA5 reanalysis variables, and CPCB ground observations.
- Captures interactions between aerosols, trace gases, meteorological parameters, and surface-level pollution.
- Improves spatial coverage and reliability, especially in areas with sparse ground monitoring.
- Enables a holistic understanding of atmospheric processes affecting air quality.

5. AI-Based Pollution Estimation

- Employs a Multi-Output XGBoost Regressor to predict six pollutants simultaneously.
- Trained on engineered features (lag, rolling, interaction, and meteorological derivatives) to model complex atmospheric relationships.
- Compared with CNN-LSTM and Random Forest models; XGBoost selected based on superior performance ($R^2 \approx 0.8445$) and efficiency.
- Offers fast inference (~17.5 ms/sample) with a compact model size (~7.85 MB).

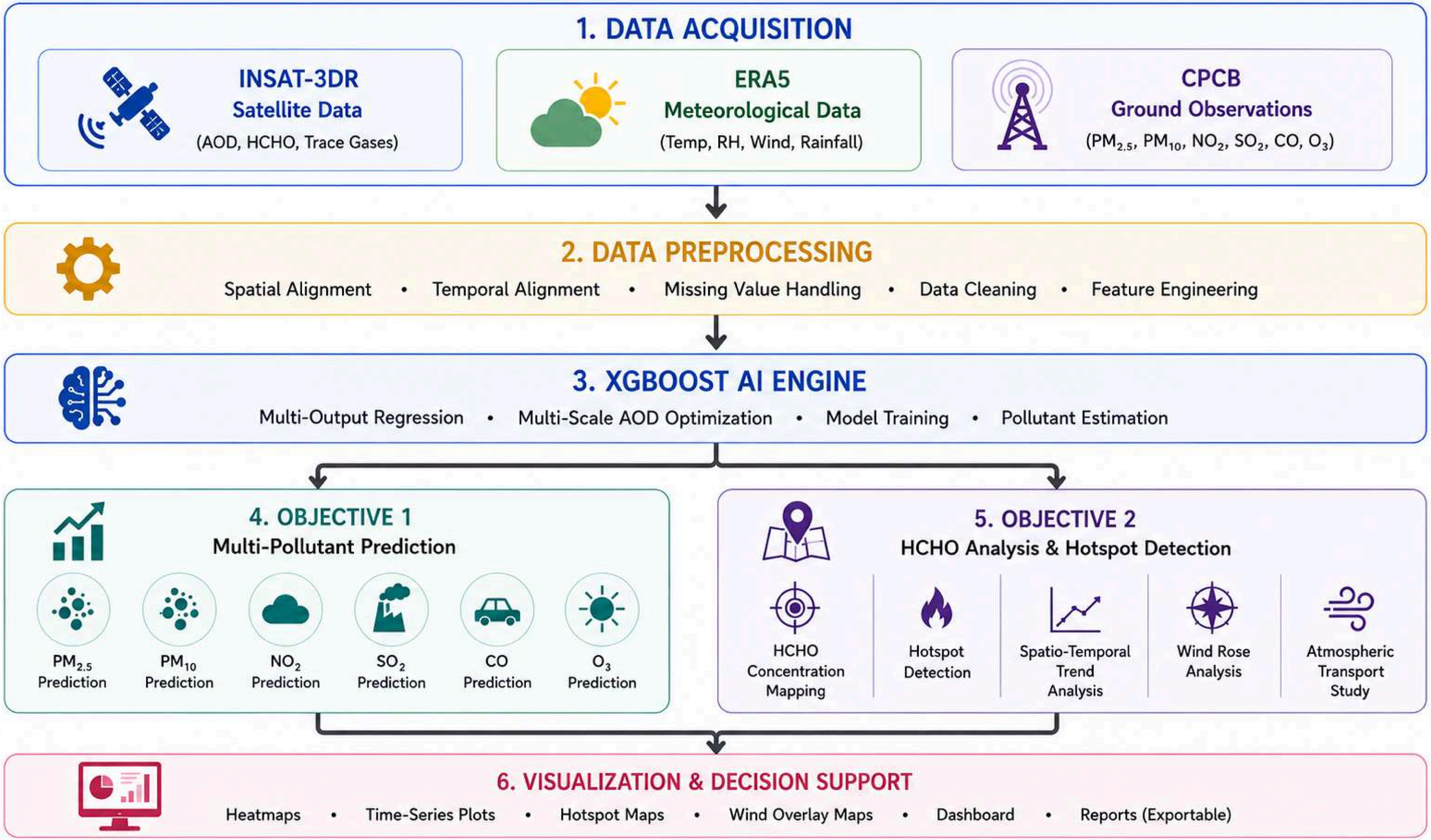
6. Atmospheric Transport & Wind Rose Analysis (Objective 2)

- Generates wind rose plots to identify prevailing wind directions and their frequency.
- Uses wind fields (u, v) and trajectory analysis (e.g., HYSPLIT style back-trajectories) to examine transport pathways and source regions.
- Analyzes long-range transport and dispersion patterns influencing HCHO distribution.
- Helps in understanding cross-boundary pollution transport and source impacts.

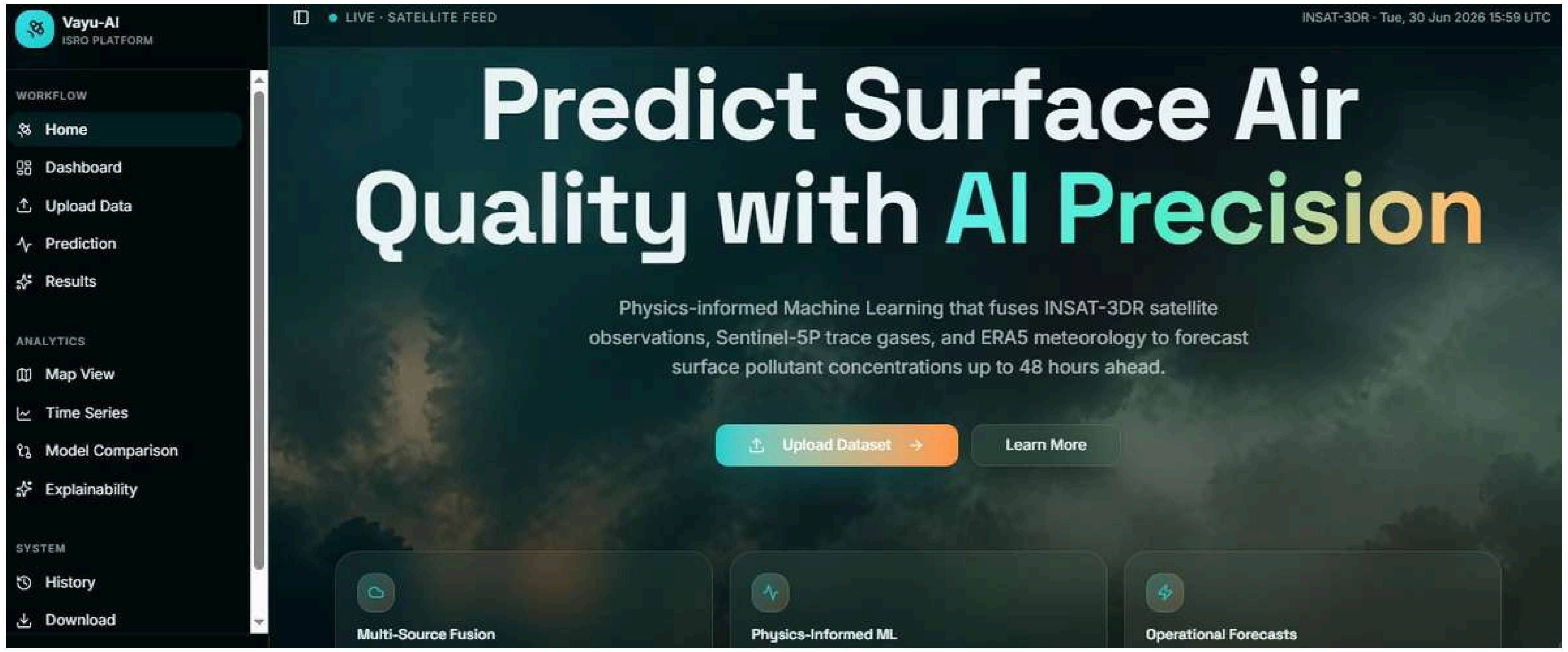
7. Spatio-Temporal Trend Analysis (Objective 2)

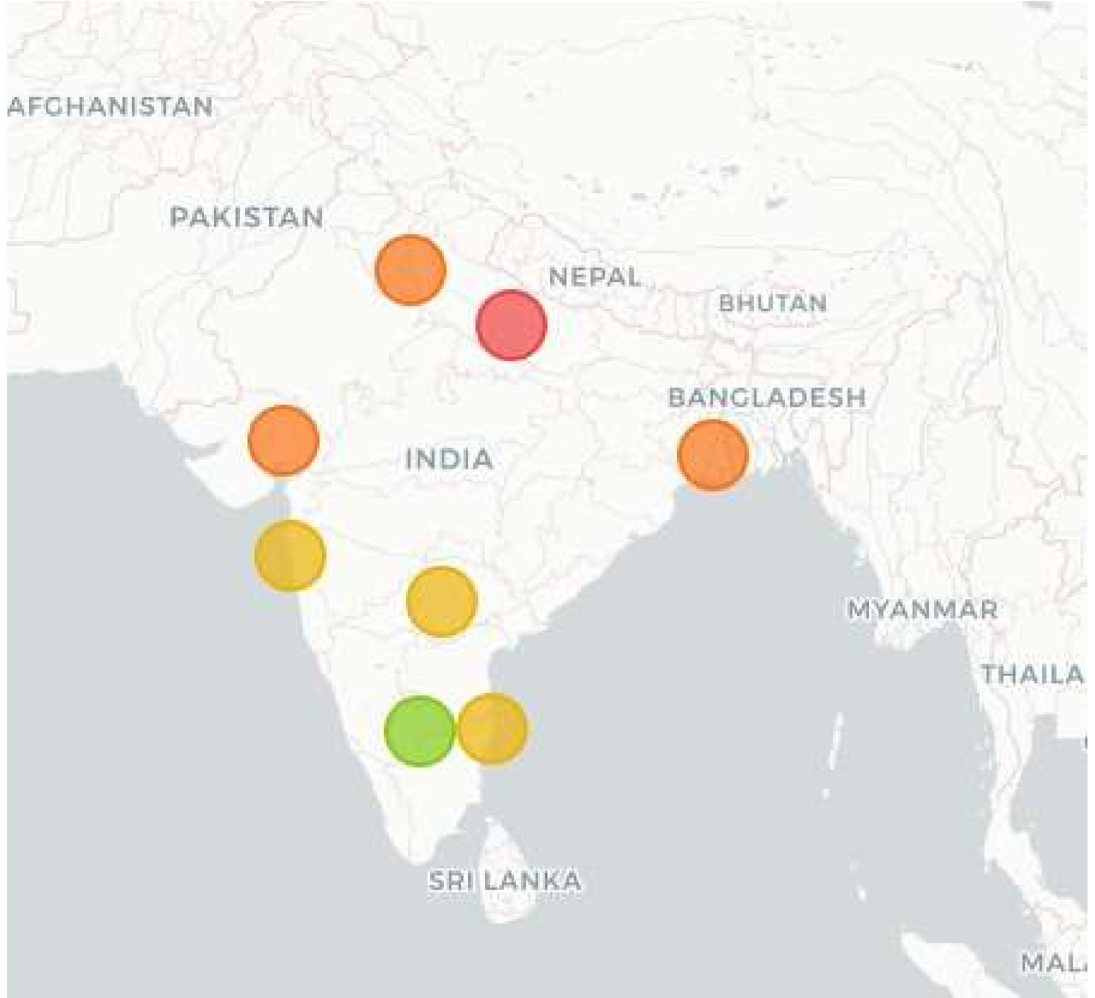
- Evaluates diurnal, weekly, monthly, and seasonal variations in HCHO concentrations and hotspot behavior.
- Produces spatio-temporal heatmaps and time-series analyses to study emission patterns and periodicity.
- Correlates trends with meteorological factors and fire counts (VIIRS / MODIS) for deeper insights.
- Supports long-term monitoring, impact assessment, and policy-making for cleaner air initiatives.

PROCESS FLOW DIAGRAM

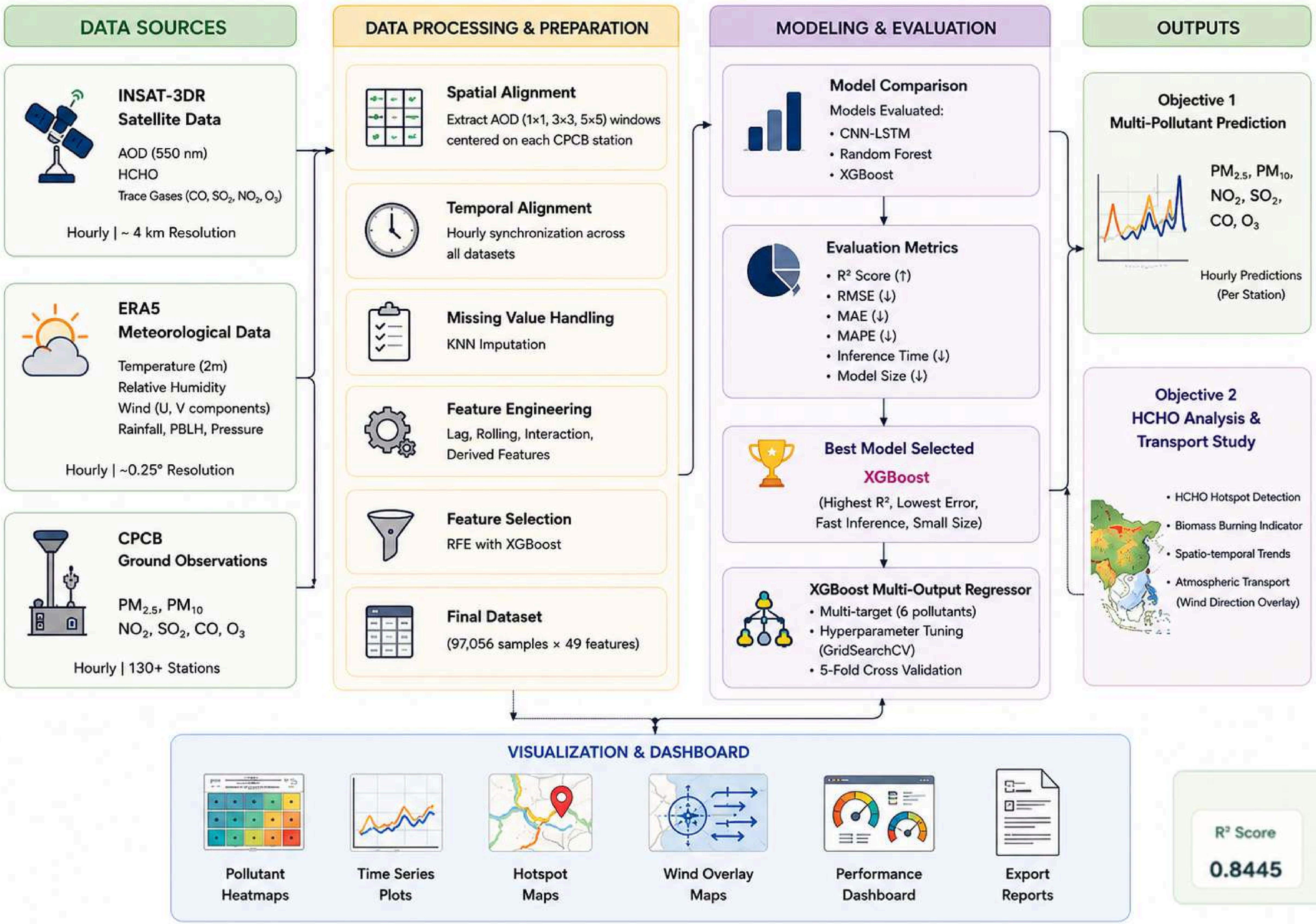


WIREFRAMES & MOCKUPS





ARCHITECTURE DIAGRAM



EXPERIMENTAL INSIGHT

- **Best Performing Model:** XGBoost consistently outperformed CNN-LSTM and Random Forest across all evaluation metrics.
- **Optimal Spatial Window:** The 5×5 INSAT-3DR AOD window provided the best predictive performance (highest R^2, lowest RMSE and MAE).
- **High Predictive Accuracy:** XGBoost achieved an average R^2 of 0.8445 with low errors (RMSE: 11.634, MAE: 7.281) for multi-pollutant prediction.
- **Efficient Inference:** Average inference time of 17.54 ms/sample, enabling real-time prediction suitability.
- **Strong Generalization:** Consistent performance across different stations and time periods with minimal overfitting.
- **Multi-Source Learning:** The fusion of INSAT-3DR, ERA5 and CPCB data significantly improved model robustness and prediction reliability.
- **Captures Complex Dependencies:** Effectively learns spatial, temporal and meteorological relationships influencing air quality.
- **Supports Dual Objectives:** Enables accurate multi-pollutant estimation and reliable HCHO hotspot detection with transport analysis.
- **Reliable & Scalable:** Lightweight model (7.85 MB) with high accuracy, suitable for large-scale deployment and decision support systems.

MODEL PERFORMANCE (TEST SET)

R^2 Score	RMSE	MAE	MAPE (%)	Inference Time	Model Size
0.8445	11.634	7.281	15.220	17.54 ms/sample	7.85 MB

TOOLS & TECHNOLOGIES

LANGUAGES

Python

AI & ML

XGBoost

Random Forest

CNN-LSTM

Scikit-learn

TensorFlow / Keras

SATELLITE & WEATHER

INSAT-3DR

ERA5

CPCB

Google Earth Engine

LIBRARIES

NumPy

Pandas

Matplotlib

Seaborn

SHAP

BHARATIYA ANTARIKSH HACKATHON 2026

THANK YOU

[Link to Research Docs]